

Introduction to Mobile IP and Addressing

Module 36

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Video: <https://www.youtube.com/watch?v=I2IzxqpP7M8>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Sunil Kumar, Department of Electrical and Computer Engineering, Punjab Agriculture University, Ludhiana

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0:00 [Music]

0:39 Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture

0:47 we will study about mobile IP and its related terms first of all please see the contents which I shall

0:55 cover first I will cover the brief introduction about mobile IP and addressing after covering brief

1:03 introduction about mobile IP and addressing I shall take up agents in Mobile IP after that I will cover the

1:10 three phases used in Mobile communication and last I shall take up inefficiency in Mobile IP so now let us

1:20 start mobile communication has received a lot of attention in last decade the interest in Mobile communication on the

1:28 internet means that the IP protocol originally designed for stationary devices must be enhanced to allow the

1:36 use of mobile computers computers that move from one network to another addressing the main problem that must be

1:43 solved in providing mobile communication using the IP protocol is addressing stationary hosts the original

1:53 Ip addressing was based on the assumption that a host is stationary attached to one specific Network a

2:02 router uses an IP address to route an IP datagram an IP address has two parts a

2:09 prefix and a suffix the prefix Associates a host to a network for example the IP address

2:18 10.3.4 24x 8 defines a host attached to the network

2:25 10.0.0.0 by 8 this implies that a host in the internet does not have an address

2:33 that it can carry with itself from one place to another the address is valid only when the host is attached to the

2:40 network if the network changes the address is no longer valid routers use this Association to

2:48 root a packet they use the prefix to deliver the packet to the network to which the host is attached this scheme

2:56 works perfectly with stationary hosts the IP addresses are designed to work with stationary hosts because part of

3:04 address defines the network to which the host is attached mobile host when a host moves from one network to another the IP

3:13 addressing structure needs to be modified several Solutions have been proposed changing the address one simple

3:22 solution is to let the mobile host change its address as it goes to the new network The Host can use

3:31 DHCP to obtain a new address to associate it with the new network this approach has several drawbacks first the

3:40 configuration files would need to be changed second each time the computer moves from one network to another it

3:47 must be rebooted third the DNS tables need to be revised so that every other host in the internet is aware of the

3:56 change fourth if the host roams from from one network to another during a transmission the data exchange will be

4:04 interrupted this is because the ports and IP addresses of the client and the server must remain constant for the

4:12 duration of the connection two addresses the approach that is more feasible is the use of two

4:19 addresses the host has its original address called the home address and a temporary address called the care of

4:27 address as per figure the home address is permanent it Associates the host to its home network the network that is

4:35 permanent home of the host the care of address is temporary when a host moves from one network to another the care of

4:44 address changes it is associated with the foreign Network the network to which the host moves mobile IP has two

4:52 addresses for a mobile host one home address and one care off address the home address is permanent the care of

5:00 address change as the mobile host moves from one network to another agents to make the change of address transparent

5:07 to rest of the internet requires a home agent and a foreign agent figure shows the position of a home agent relative to

5:15 the home network and a foreign agent relative to foreign Network we have shown the home and the foreign agent as routers but we need to emphasiz that the

5:24 specific functions as an agent is performed in the application layer in other words they are both routers and

5:31 host home agent the home agent is usually a router attached to the home network of the mobile host the home

5:40 agent acts on behalf of the mobile host when a remote host sends a packet to the mobile host the home agent receives the

5:49 packet and sends it to the foreign agent foreign agent the foreign agent is usually a router attached to the foreign

5:58 Network the foreign agent receives and delivers package sent by the Home agent to the mobile host the mobile host can

6:06 also act as a foreign agent in other words the mobile host and foreign agent can be the same however to do this a

6:14 mobile host must be able to receive a care of address by itself which can be done through using the

6:22 DHCP in addition the mobile host needs the necessary software to allow it to communicate with the home agent and to

6:30 have two addresses its home address and its care of address this dual addressing must be transparent to the application

6:38 programs when the mobile host acts as a foreign agent the care of address is called a colocated care of address when

6:48 the mobile host and the foreign agent are the same the care of address is called a collocated care of

6:56 address the advantage of using a c ated care of addresses that the mobile host can move to any network without worrying

7:05 about the availability of a foreign agent three phases in Mobile communication to communicate with a

7:13 remote host a mobile host goes through three phases agent Discovery registration and data transfer as shown

7:22 in the figure the first phase agent Discovery involves the mobile host the foreign agent and the home

7:31 agent the second phase registration also involves the mobile host and the two

7:38 agents finally in the third phase the remote host is also involved we discuss each phase

7:47 separately agent Discovery the first phase in Mobile communication agent Discovery consists of two surfaces a

7:56 mobile host must discover a home agent before it leaves its home network a mobile host must also discover a foreign

8:04 agent after it has moved to a foreign Network this discovery consists of learning the care of address as well as

8:13 the foreign agent address the discovery involves two types of messages advertisement and

8:20 solicitation agent advertisement when a router advertises its presence on a network using an icmp

8:28 router advertisement it can append an agent advertisement to the packet if it acts as an agent figure shows how an

8:38 agent advertisement is piggy bagged to the router advertisement packet the field descriptions shown in figure are

8:45 as follows type the 8bit type field is set to 16 length the 8bit length field defines

8:56 the total length of the extension message not the length of the icmp advertisement

9:03 message sequence number the 16bit sequence number field holds the message

9:09 number the recipient can use the sequence number to determine if a message is

9:16 lost lifetime the lifetime field defines the number of seconds that the agent will accept

9:24 requests if the value is a string of ones the lifetime is in finite code the

9:31 code field is an 8bit flag in which each bit is set one or unset to

9:39 zero care of addresses this field contains a list of addresses available

9:45 for use as care of addresses the mobile host can choose one of these addresses

9:52 the selection of this SC of address is announced in the registration request note that this field is used only by a

10:00 foreign agent agent solicitation when a mobile host has

10:07 moved to a new network and has not received agent advertisements it can initiate an agent

10:14 solicitation it can use the icmp solicitation message to inform an agent that it needs

10:22 assistance registration the second phase in Mobile communication is registration after a

10:30 mobile host has moved to a foreign Network and discovered the foreign agent it must register there are four aspects

10:38 of registration first the mobile host must register itself with the foreign agent second the mobile host must

10:47 register itself with its home agent this is normally done by the foreign agent on behalf of the mobile host third the

10:55 mobile host must renew registration if it has expired fourth the mobile host must cancel its registration when it

11:03 returns home request and reply to register with the foreign agent and the home agent the mobile host uses a

11:11 registration request and a registration reply registration request a registration request is sent from the

11:20 mobile host to the foreign agent to register its care of address and also to

announce its home address and home agent

- 11:28 address the foreign agent after receiving and registering the request relays the message to the home agent note that the
- 11:37 home agent now knows the address of foreign agent because the IP packet that is used for relaying has the IP address
- 11:45 of the foreign agent as the source address figure shows the format of the registration request the field
- 11:53 descriptions shown in figure are as follows type the the 8bit type field defines the type of message for a
- 12:02 request message the value of this field is one flag the 8bit flag field defines
- 12:09 forwarding information the value of each bit can be set or unset lifetime this
- 12:17 field defines the number of seconds the registration is valid if the field is a string of zeros the request message is
- 12:25 asking for D registration if the field is a string of once the lifetime is
- 12:32 infinite home address this field contains the permanent address of the mobile host mobile agent address this
- 12:41 field contains the address of the home agent care of address this field is the
- 12:48 temporary address of mobile host identification this field contains the 64bit number that is inserted into the
- 12:56 request by the mobile host and repeat in the reply message it matches a request with a reply
- 13:05 extensions variable length extensions are used for authentication they allow a home agent to authenticate the mobile
- 13:13 agent registration reply a registration reply is sent from the home agent to the foreign agent and then relay to the
- 13:22 mobile host the reply confirms or denies the registration request data transfer after agent
- 13:31 Discovery and registration a mobile host can communicate with the remote host figure shows the process of data
- 13:39 transfer I'll explain one by one from remote host to home
- 13:45 agent path one of figure shows this step when a remote host wants to send a packet to the mobile host it uses its
- 13:54 address as the source address and the home address of the mobile host as the destination address in other words the
- 14:02 remote host sends a packet as though the mobile host is at its home network the packet however is intercepted by the
- 14:10 Home agent which pretends it is the mobile Host this is done using the proxy ARP
- 14:18 technique from home agent to foreign agent path two of the figures shows the
- 14:25 step after receiving the packet the home agent sends the packet to the foreign

agent using the tunneling concept the

14:34 home agent encapsulates the whole IP packet inside another IP packet using its address as the source and the

14:42 foreign agent's address as the destination from foreign agent to mobile host part three of figure shows this step

14:51 when the foreign agent receives the packet it removes the original packet however since the destination address is

14:59 is the home address of mobile host the foreign agent consults a registry table to find the care of address of the

15:06 mobile host otherwise the packet would just be sent back to the home network the packet is then sent to the care of

15:14 address from mobile host to remote host path four of figure shows the step when a mobile host wants to send a packet to

15:23 a remote host for example a response to the packet it has received it sends as it does normally the mobile host

15:31 prepares a packet with its home address as the source and the address of remote host as the destination although the packet comes

15:40 from the foreign Network it has the home address of the mobile host transparency in this data transfer process the remote

15:49 host is unaware of any Movement by the mobile host the remote host send packets using the home address of the mobile

15:56 host as the destination address it receives the packets that have the home address of the mobile host as the source

16:04 address the movement is totally transparent the rest of the internet is not aware of the mobility of mobile

16:12 host inefficiency in Mobile IP communication involving mobile IP can be

16:20 inefficient the inefficiency can be severe or moderate this severe case is called double crossing or 2X

16:29 the moderate case is called triangle routing or dog leg routing double crossing according to

16:37 figure double crossing occurs when a remote host communicates with the mobile host that has moved to the same network

16:46 as the remote host when the mobile host sends a packet to the remote host there is no inefficiency the communication is

16:55 local however when the remote host sends a packet to the mobile host the packet crosses the internet twice since a

17:03 computer usually communicates with other local computers the inefficiency from double crossing is significant triangle

17:11 routing triangle routing occurs when the remote host Communications with the mobile host that is not attached to the

17:19 same network or site as the mobile host when the mobile host sends a packet to the remote host there is no inefficiency

17:29 however when the remote host sends a packet to the mobile host the packet goes from remote host to the home agent

17:36 and then to the mobile host the packet travels the two sides of a triangle instead of just one side one solution to

17:45 inefficiency is for the remote host to bind the care of address to the home address of a mobile host for example

17:53 when a home agent receives the first packet for a mobile host it forwards the packet to the foreign agent it could

18:01 18 minutes, 1 second also send an update binding packet to the remote host so that future packets to this host can be sent to the care of

18:09 address the remote host can keep this information in a cache the problem with this strategy is that the cash entry

18:18 becomes outdated once the mobile host Moves In this case the home agent needs to send a warning packet to the remote

18:26 host to inform it of the change so friends this was all about mobile IP and its related terms hope the concepts

18:34 explained in this lecture were understandable and helpful hope to see you in next lecture till then goodbye thank you

18:44 [Music]

19:10 [Music]

19:59 [Music]